

B. NIHANTH

Java Backend Developer | AI Engineer | Spring Boot | Microservices | AWS
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PROFESSIONAL SUMMARY

Results-driven Java Backend Developer and AI Engineer with hands-on experience building scalable microservices architectures, high-performance REST APIs, and intelligent autonomous systems. Proficient in Spring Boot, Hibernate, AWS, and Docker, with a strong foundation in distributed systems and system design. Complements backend expertise with applied AI and ML capabilities — from generative AI pipelines to autonomous vehicle systems. Published patent holder, AWS certified, and organiser of 40+ technical events. Passionate about engineering robust, intelligent systems that solve real-world problems at scale.

PROJECTS

Salvii AI — Scalable AI Web Assistant

2025

Spring Boot · Generative AI · REST APIs · React.js · Performance Optimisation

- Built a scalable AI-powered web application for intelligent content generation and problem solving using generative AI integration.
- Optimised API pipeline and response caching strategies, reducing end-to-end response latency by 25%.

Smart Autonomous EV Delivery System

2025

Java · Spring Boot · Microservices · React.js · MongoDB · CAN Protocol · AWS

- Architected an AI-powered autonomous vehicle delivery system with optimised routing algorithms, improving route efficiency by 30%.
- Implemented CAN protocol communication enabling real-time data exchange between vehicle subsystems with sub-millisecond latency.
- Developed a microservices-based monitoring dashboard processing real-time telemetry data using React.js and MongoDB.

Online Certification Management System

2024

Java · Spring Boot · React.js · Hibernate ORM · MySQL · GitHub Actions · CI/CD

- Engineered a scalable certification management platform reducing administrative workload by 40% and system efficiency by 35%.
- Designed secure REST APIs for certificate issuance and validation; optimised Hibernate ORM queries reducing DB response time by 25%.
- Automated deployment pipelines using GitHub Actions and CI/CD, cutting release cycles by 40%.

FindMyItem — Lost Item Recovery Platform

2025

Spring Boot · React.js · REST APIs · Search Algorithms · Full Stack

- Developed a full-stack lost-and-found platform serving 500+ students, improving item recovery efficiency by 35%.
- Implemented categorised search and filtering algorithms reducing item lookup time by 40%; built performant RESTful APIs with Spring Boot.

RESEARCH

Green Hydrogen Generation using Nanomaterials and Machine Learning

Published Patent — IPASS India

- Proposed AEM-based hydrogen production using quaternary nanomaterials with ML models to predict catalytic efficiency and optimise material performance.
- Applied AI-driven optimisation techniques to improve experimental accuracy and material analysis — contributing to sustainable energy research.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

2023 – 2027

Koneru Lakshmaiah University (KL University), Vijayawada

CGPA: 9.5 / 10.0

TECHNICAL SKILLS

Languages Java, C++, C, JavaScript

Backend Spring Boot, REST APIs, Microservices, Hibernate ORM, JPA, System Design

Frontend React.js, Node.js, HTML5, CSS3

AI / ML Machine Learning, Deep Learning, Generative AI, LLM Integration

Databases MySQL, PostgreSQL, MongoDB

Cloud / DevOps AWS, Docker, GitHub Actions, CI/CD, Git, Linux

Tools IntelliJ IDEA, VS Code, Postman, Prometheus, Grafana

Concepts Distributed Systems, API Design, Data Structures, Algorithms

CERTIFICATIONS

AWS Certified Cloud Practitioner | Oracle Cloud Infrastructure Foundations Associate | Salesforce AI Associate | Frontend Developer (React.js) | Java Programming | Problem Solving

LEADERSHIP & ACHIEVEMENTS

- Organised 40+ technical and research events, hackathons, and workshops — increasing student participation by 45%.
- Participated in national hackathons including IIT Madras Shaastra AI Hackathon.
- Published patent on Green Hydrogen Generation using Nanomaterials and Machine Learning — IPASS India.
- Maintained CGPA of 9.5/10 while concurrently managing multiple live production projects and research work.